

A 0.1% L-Menthol Mouth Swill in Elite Male Rugby Players Has Different Effects in Forwards and Backs

Marcia L. Jerram,¹ Dane Baker,¹ Tiaki B. Smith,² Phil Healey,³ Lee Taylor,^{4,5,6} and Katherine Black¹

¹Department of Human Nutrition, University of Otago, Dunedin, New Zealand; ²University of Waikato, Hamilton, New Zealand; ³Blues Super Rugby Franchise, Auckland, New Zealand; ⁴School of Sport, Exercise and Health Sciences, National Centre for Sport and Exercise Medicine (NCSEM), Loughborough University, Loughborough, United Kingdom; ⁵Human Performance Research Centre, University of Technology Sydney (UTS), Sydney, NSW, Australia; ⁶Sport & Exercise Discipline Group, Faculty of Health, University of Technology Sydney (UTS), Sydney, NSW, Australia

Purpose: Menthol mouth swills can improve endurance performance in the heat, which is attributed to attenuations in nonthermally derived thermal sensation (TS) and perception of effort. However, research in elite team-sport athletes is absent. Therefore, this study investigated the performance and TS responses to a 0.1% menthol mouth rinse (MR) or placebo (PLA) among elite male rugby union players. **Method:** Twenty-seven (15 Forwards and 12 Backs) elite male Super Rugby players completed two 3-minute 15-a-side rugby-specific conditioning blocks, with MR or PLA provided at the start of training (baseline), at the start of each 3-minute block (swill 1 [S1] and swill 2 [S2]), and at the end of training (swill 3 [S3]). TS was assessed using the American Society of Heating, Refrigerating and Air-Conditioning Engineers 9-point Analog Sensation Scale after each swill and at baseline (preconditioning block). Acceptability was measured after baseline swill and S3 using a 5-question Likert scale. Physical performance was measured throughout training using global positioning system metrics. **Results:** MR attenuated TS from baseline to S1 ($P = .003$, $SD = 1.01$) and S2 ($P = .002$, $SD = 1.09$) in Forwards only, compared with PLA. Acceptability was higher only for Forwards in MR versus PLA at baseline ($P = .003$, $SD = 1.3$) and S3 ($P = .004$, $SD = 0.75$). MR had no effect on physical performance metrics ($P > .05$). **Conclusion:** MR attenuated the rise in TS with higher acceptability at S1 and S3 (in Forwards only) with no effect on selected physical performance metrics. Longer-duration exercise (eg, a match) in hot-humid conditions eliciting markedly increased body temperatures could theoretically allow favorable changes in TS to enhance performance—these postulations warrant experimental investigation.

Keywords: team sport, thermal sensation, body composition

Rugby union is an intermittent, high-intensity team sport, even though ~85% of game time is spent at low intensities.¹ These high-intensity activities such as repeated sprints and explosive movements (eg, tackles, rucks, mauls) stress players' anaerobic capacity.¹ Demands differ by position, with Forwards spending more time in rucks and mauls (~25% more than backs) and Backs completing 2 to 3 times more high-intensity running than Forwards.² Backs cover greater distances (7227 m vs 6680 m in Forwards) and spend more time "sprinting" > 20 km·h⁻¹ (524 m vs 313 m in Forwards), while Forwards experience more impacts (1274 vs 798 in Backs) and lower maximum rest periods (3.4 min vs 6.4 min in Backs).^{2,3}

Exercising in the heat can raise core body temperature and cause significant feelings of discomfort, all which can impair performance by reducing time to fatigue (physical and cognitive), compromising the ability to sustain high-intensity efforts required in match play.⁴⁻⁶ This in turn can lead to a greater susceptibility of exertional heat illnesses/stroke.⁷ World Rugby Sevens Tournament match play found that several players reached a core temperature (T_c) $\geq 39^\circ\text{C}$ with as little as ~7 minutes on pitch in temperate/mild conditions.^{8,9} It is likely higher body temperatures would be seen in the 15-a-side game due to the prolonged physical demands outlined

above, executed within warm or hot-humid environmental conditions (however, no such T_c data are available for 15-a-side rugby union). For context, $T_c \geq 39^\circ\text{C}$ can impair intermittent sprint performance, an evidently important physical quality within rugby union.⁴ Rugby union players experience fluid losses of 1.74 (0.32) L h⁻¹ during training¹⁰ while in match play, values associated with significant thermoregulatory strain (Forwards: 1.94 [0.14] kg; Backs 1.04 [0.17] kg) are seen.¹¹ Body composition also can contribute to thermoregulatory responses during exercise dependent on adiposity and size.¹² Forwards generally have a relatively higher body mass and percentage body fat (%body fat) than Backs^{1,13} with these differences capable of differentially modifying thermoregulation and heat stress.^{14,15}

Heat acclimation/acclimatization is the gold standard strategy to protect athletes against performance decrements and exertional heat illnesses/stroke.⁵ However, various logistical (travel, COVID-19 restrictions, fixture congestion, etc) and practical (facility access, home environment, injury, etc) challenges may preclude its integration within the complex physical preparation and recovery schedules of Super Rugby players. Hence, there are occasions, where complimentary to (desirable) or in place of (not desirable) heat acclimation/acclimatization, preexercise, and during exercise cooling interventions may be desirable.¹⁶ Physical cooling interventions resulting in negative heat exchange may include the use of cooling vests, ice slushies, facial sprays, and cold-water immersion.⁷ Perceptual cooling strategies, including the use of menthol mouth rinsing, have gained popularity with practitioners and athletes as they are quick and easy to administer.^{7,17} Subsequently, research has shown physical performance may be altered through

Baker  <https://orcid.org/0000-0001-7406-0709>

Smith  <https://orcid.org/0000-0003-3762-6253>

Taylor  <https://orcid.org/0000-0002-8483-7187>

Black (katherine.black@otago.ac.nz) is corresponding author,  <https://orcid.org/0000-0003-4672-3362>

interventions, including menthol,^{16,18–20} that target the central nervous system to manipulate perception during exercise.^{17,21,22} Menthol, specifically the L-isomer form (L-menthol) can be swilled around the oral cavity, activating gustatory, and temperature receptors,^{23,24} triggering the transient receptor potential melastatin 8 cold receptor/channels,^{23,24} evoking a sensation of coolness and freshness in the mouth and nasal passages.^{18,25} Evidence surrounding the use of menthol to attenuate thermal comfort/sensation^{19,26} and improve endurance performance has been published.^{16,19} Its effect in high-intensity, intermittent sports such as rugby union is weaker with some data indicating that even with improved thermal comfort, no performance benefits are evident.²⁵ However, the majority of menthol work did not utilize elite athletes, who are known to have a modified perception of pain (and likely temperature sensation, given that they are chronically partially heat acclimated)²¹ compared with nonelite athletes.

Thus, the need for research in field-based, “real world” conditions on elite athletes⁵ is warranted to understand the effectiveness and to consider strategies for the use of a L-menthol mouth rinse (MR), which may differ from research conducted in laboratory-based settings in nonelite athletes.²¹ The main aim of the current study was to investigate the use of a L-menthol MR swill on the performance, thermal sensation (TS), and acceptability in elite rugby players during a rugby-specific conditioning block. It is hypothesized that (1) a menthol mouth swill will benefit performance during rugby-specific conditioning and (2) menthol will attenuate the rise in TS, which could differ between Forwards and Backs.

Methods

Subjects

Thirty professional male rugby players from a New Zealand Super Rugby franchise (the highest rugby competition in New Zealand), aged 24.2 (3.2) years were recruited for this study. Of the 30 players, 11 were Samoan, 8 Māori, 9 New Zealand European or European, and 2 Fijian. The data from 3 participants were excluded from the final data set, due to injury or missing data. Thus, 27 participants were included in the final analysis. There were 15 Forwards and 12 Backs, the Forwards were heavier than the Backs (Forwards 115.32 [8.75] kg, Backs 97.24 [8.20] kg) and had higher $\Sigma 8$ skinfolds (Forwards 83.06 [16.04] mm, Backs 64.24 [11.30] mm). The study was conducted during preseason training, and athletes were instructed to carry out normal training procedures as instructed by coaching staff throughout testing. Athletes were familiarized with the experimental protocol, which was designed to minimize disruptions to training. All athletes were injury free and completed at least 10 training sessions per week, which included rugby-specific conditioning, skills, and gym (strength and power) training. The study was conducted over 2, team field training sessions (trials 1 and 2) separated by 3 days.

Design

This was a randomized, crossover study to investigate the effects of a 0.1% L-menthol mouth swill on performance, acceptability, and TS in elite rugby players. Ethical approval was obtained from the University of Otago Human Ethics Committee (19/167). Participants were informed of the experimental protocol and provided informed written consent in the spirit of the Declaration of Helsinki prior to any measures being taken.

Body-Composition Measures

Prior to experimental testing, skinfold assessment was obtained by a certified International Society for the Advancement of Kinanthropometry-accredited anthropometrist. Measures were taken from the biceps, triceps, subscapular, iliac crest, supraspinale, abdominal, thigh, and calf. All of the measures were obtained using skinfold calipers and accumulated to provide a $\Sigma 8$ skinfold measure; percentage of body fat was calculated using the Yuhasz formula.²⁷

Menthol and Placebo Solution Formulation and Description

A 0.1% concentration L-menthol solution was prepared on the morning of the experimental protocol by dissolving L-menthol crystals (pharmaceutical grade, Sartorius) in boiling tap water.^{19,28,29} The L-menthol crystals were independently batch tested according to the guidelines for safe supplement use by the Banned Substances Control Group. The solution was then stored in a freezer to be chilled. Once chilled, 25 mL of the solution was decanted into sample containers (Salter). The 25-mL placebo solution was standard tap water and was stored in sample containers in the refrigerator alongside the L-menthol. Menthol and placebo samples were transported to a second refrigerator adjacent to the rugby field, approximately 1 hour before testing. The solutions remained there until required. Solution temperature (in degree Celsius) was recorded using a food-grade digital thermometer (Grillman). A temperature of 8 °C was recorded prior to the session and 15 °C at the completion of the session. The athletes were blinded to which solution they were given; however, absolute blinding was not attained as the athletes were able to distinguish (via taste and mouth feel) what solution they had received when completing the mouth swill.

Methodology

Both testing sessions consisted of a normal warm-up, unit session consisting of position-specific tactical plays, skills, as well as a contact conditioning block, before commencing a rugby-specific conditioning session. A detailed overall description of the training session is shown in Figure 1. Players provided a baseline swill measure of TS followed by either a MR or PLA and an acceptability questionnaire. Players then completed two 3-minute (swill 1 [S1] and swill 2[S2]), 15-a-side rugby-specific conditioning blocks (6 min in total) with a TS questionnaire, followed by an MR or PLA provided at the start of each 3-minute block (S1 and S2). At the completion of the conditioning blocks, normal training resumed (posttest) which consisted of; an additional rugby-specific conditioning block and a “work-on” block, which varied slightly between trials, and was therefore not included in the final analysis. Trial 1 posttest mean duration was 33 minutes and 23 seconds, and trial 2 posttest mean duration was 29 minutes and 41 seconds. At the end of training (S3) either a MR or PLA MR was provided, followed by an acceptability questionnaire (any comments were recorded here). The body weights of the players were recorded again at the completion of training, implementing the same protocol used at the beginning of the session.

Fluid Balance. Each player was provided with their own-named drink bottle and instructed to only drink from their individual bottles. Powerade and water were provided throughout the training period. However, when the mouth swill was administered only water was provided. All drink bottle weights were recorded pretraining and posttraining. Throughout the training session, if necessary, bottle weights were recorded when empty to determine how much had been consumed and once again when they had been refilled (Salter). Body

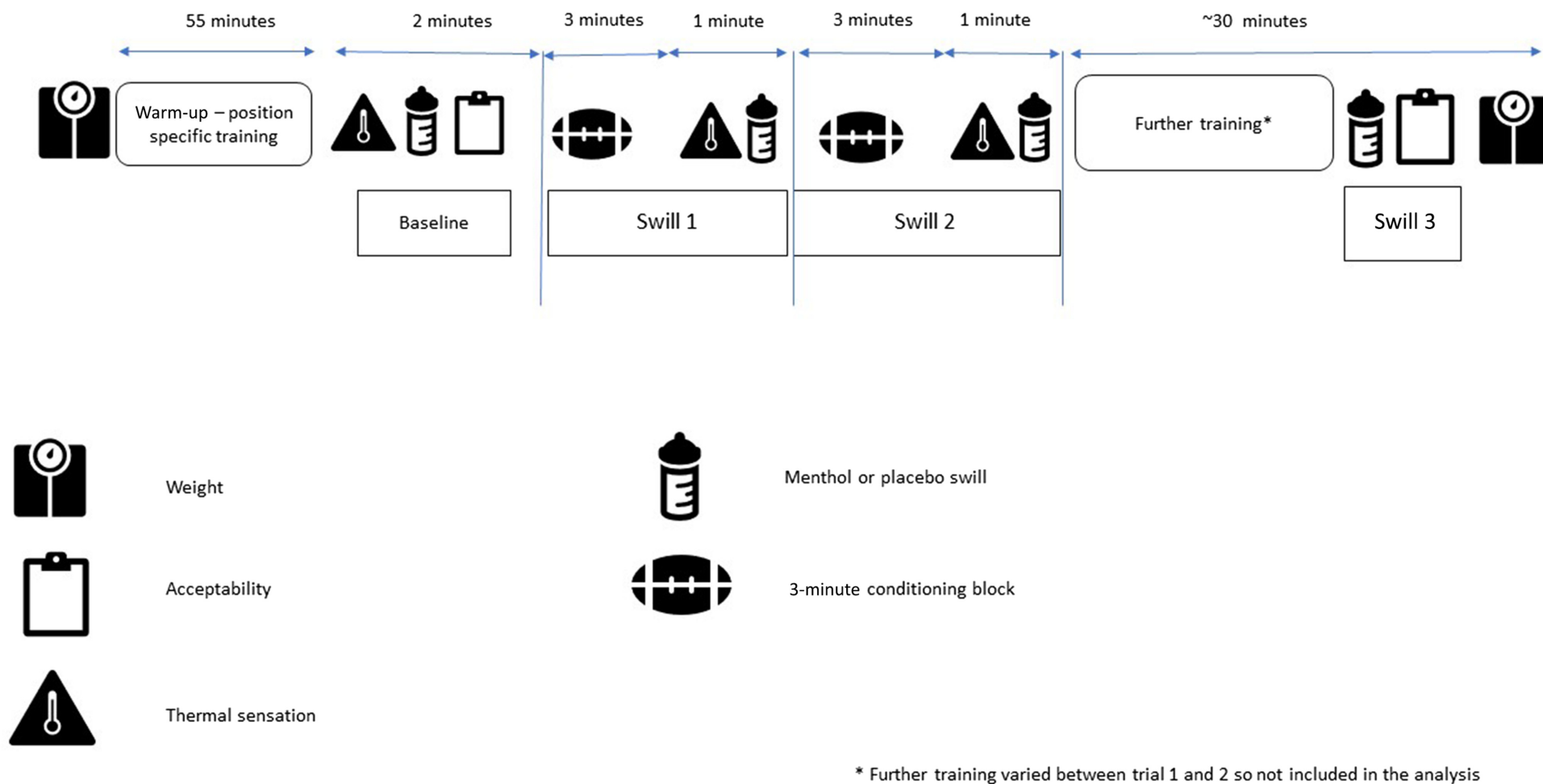


Figure 1 — Schematic of study protocol.

mass (Dolphin) and fluid intake data were obtained pretraining and posttraining to determine fluid gains and losses over the training session. Sweat loss (in liters) was determined by calculating the difference between player prebody and postbody weight, accounting for fluid intake during the training session. From this, sweat rate (in liter per hour) was calculated by dividing the total sweat loss per player by the total length of the session. The percentage dehydration was also calculated by determining the percentage difference between player prebody and postbody weight.

TS and Acceptability Data. The TS was recorded using the American Society of Heating, Refrigerating and Air-Conditioning Engineers 9-point Analog Sensation Scale ($-4 = \text{“very cold,”}$ $0 = \text{“neutral,”}$ and $+4 = \text{“very hot”}$ ³⁰). The acceptability questionnaire consisted of 5 questions (taste, flavor, sensation, aftertaste, and overall pleasantness) and included a Likert scale ($1 = \text{dislike extremely,}$ $5 = \text{neither like nor dislike,}$ $10 = \text{like extremely}$).

Global Positioning System Data. Each athlete was equipped with a global positioning system (Apex STATSports) tracker fitted into their rugby jersey prior to the training session. Workload measurements (high metabolic load/minutes [$> 25 \text{ w kg}^{-1}$] [in meters per minute], sprint distance/minutes [$> 5.5 \text{ m}\cdot\text{s}^{-1}$] [in meters per minute], high-speed running/minutes [$> 7.5 \text{ m}\cdot\text{s}^{-1}$] [in meters per minute], and total distance covered/minutes [in meters per minute]) were tracked throughout the training session, and experimental testing data were isolated to observe workload for each part of the training session.

Ambient Temperature. Ambient temperature (relative air density, wet bulb, dew point, heat index, humidity, and temperature) were recorded at the beginning, preexperimental testing protocol, postexperimental testing protocol, and at the end of the entire training session in Table 1 (Kestrel 5100).

Table 1 Ambient Conditions for Trials 1 and 2, Mean (SD)

	Trial 1	Trial 2
Ambient temperature, °C	27.7 (1.4)	25.9 (1.7)
Relative humidity, %	55.7 (1.2)	60.6 (2.2)

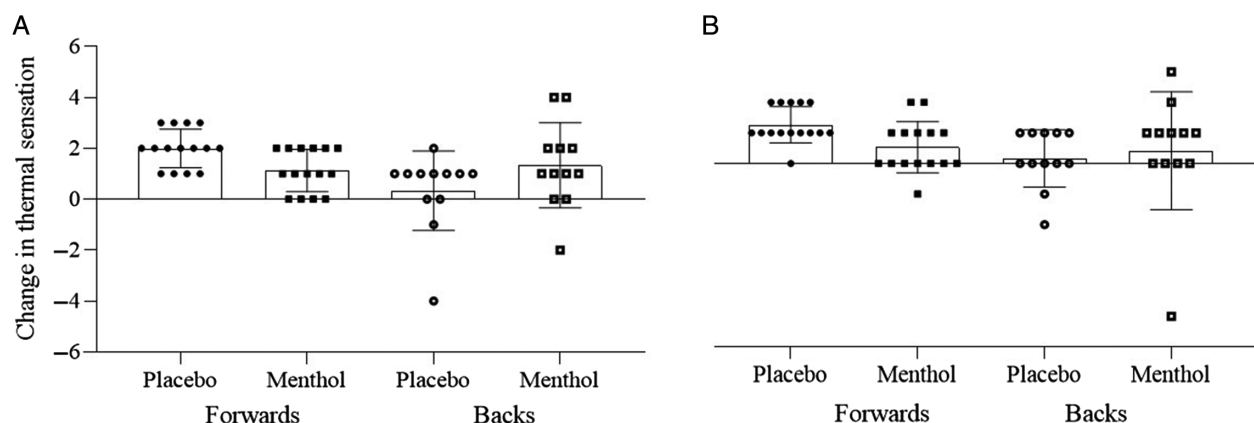


Figure 2 — Change in thermal-sensation scores for MR (round) and PLA (squares) trials (A) from baseline to swill 1 and (B) from baseline to swill 2 for backs (enclosed) and forwards (open).

Statistical Analysis

Independent samples *t* test (ie, between-subjects *t* test) and standardized differences were used to determine both the probability of any significant differences and the magnitude of any possible differences for each measure in response to a menthol mouth swill for each trial. The independent sample *t* tests were conducted in SPSS, and normality was tested using tests of kurtosis and sphericity. As change in TS data was not normally distributed, Wilcoxon signed-rank tests were performed. The alpha level used was $P \leq .05$. Data are reported as means (SD) unless otherwise stated.

Magnitudes of the standardized effects were calculated and interpreted using thresholds of < 0.2 , 0.2 , 0.6 , 1.2 , 2.0 , and 4.0 for trivial, small, moderate, large, very large, and extremely large, respectively.²⁰

Results

Thermal Sensation

The MR attenuated TS from baseline to S1 ($P = .003$, $SD = 1.01$) and S2 ($P = .002$, $SD = 1.09$) in Forwards only, compared with PLA (Figure 2A and 2B). However, for the Backs, there was no significant attenuation on TS from baseline to post S1 between MR and PLA ($P = .204$, Wilcoxon signed-rank test) nor from baseline to post S2 ($P = .375$, Wilcoxon signed-rank test). The percentage body fat and change in TS were positively associated from pre to post S2 for the PLA trial $P = .04$, $r^2 = .181$ ($n = 23$), this was not seen with the MR trial $P = .282$, $r^2 = .054$ ($n = 23$).

Acceptability Data

Acceptability was significantly higher for Forwards in MR versus PLA at baseline ($P = .003$, $SD = 1.3$) and S3 (posttest) ($P = .004$, $SD = 0.75$). This was significantly higher for Backs in MR versus PLA at baseline ($P = .028$; Table 2).

Performance

Performance metrics showed a large variability for both groups in both trials. MR had no effect on physical performance metrics ($P > .05$; Table 3).

Table 2 Acceptability Data for the Forwards and Backs Pretest and Posttest With Menthol and Placebo

	Forwards (n = 15)					Backs (n = 12)				
	Taste	Flavor	Sensation	Aftertaste	Overall	Taste	Flavor	Sensation	Aftertaste	Overall
Pretest (baseline)										
Placebo	6.1 (1.8)	5.9 (1.3)	4.9 (0.7)	5.7 (1.4)	6.1 (1.5)	5.7 (1.2)	5.7 (1.1)	5.3 (0.8)	5.4 (0.8)	5.8 (1.3)
Menthol	5.9 (1.7)	5.4 (1.5)	7.0 (2.1)	5.6 (1.8)	6.2 (2.0)	5.9 (1.6)	5.5 (2.1)	6.3 (1.5)	4.8 (1.6)	6.4 (1.7)
Standardized difference	−0.10 (trivial)	−0.38 (small)	1.30 (large)	−0.06 (trivial)	0.09 (trivial)	0.19 (trivial)	−0.12 (trivial)	0.79 (moderate)	−0.49 (small)	0.36 (small)
<i>P</i>	.399	.123	.003*	.426	.400	.243	.278	.028*	.472	.180
Posttest (swill 3)										
Placebo	7.2 (2.2)	6.9 (1.9)	5.9 (2.2)	6.8 (3.1)	6.2 (2.5)	7.1 (2.3)	6.5 (2.0)	6.6 (2.0)	6.6 (2.1)	6.5 (2.2)
Menthol	6.7 (2.4)	6.5 (2.3)	7.5 (1.9)	6.8 (3.2)	7.5 (2.2)	6.0 (3.1)	6.0 (3.2)	6.9 (2.2)	6.1 (3.4)	6.6 (2.9)
Standardized difference	−0.24 (small)	−0.18 (trivial)	0.75 (moderate)	0.00 (trivial)	0.53 (small)	−0.38 (small)	−0.17 (trivial)	0.14 (trivial)	−0.18 (trivial)	0.05 (trivial)
<i>P</i>	.203	.278	.004*	.500	.044*	.125	.234	.289	.225	.411

*Significant difference represented as $P < .05$. Bold values are statistically significant.

Table 3 GPS Measures for the Forwards and Backs With a Menthol and Placebo Mouth Swill

	Forwards (n = 15)				Backs (n = 12)			
	HSR/min, m·min ⁻¹	Sprint distance/min, m·min ⁻¹	HiMets/min	Distance/min, m·min ⁻¹	HSR/min	Sprint distance/min, m·min ⁻¹	HiMets/min	Distance/min, m·min ⁻¹
Warm-up								
Placebo	0.6 (0.5)	50.3 (103.4)	3.0 (1.0)	31.6 (4.8)	2.7 (0.7)	134.8 (179.4)	7.3 (1.3)	48.6 (5.7)
Menthol	0.8 (0.6)	50.2 (124.7)	3.6 (1.5)	34.2 (7.9)	2.8 (1.0)	148.8 (254.9)	7.3 (1.5)	49.9 (6.3)
Standardized difference	0.40 (small)	0.00 (trivial)	0.49 (small)	0.4 (small)	0.06 (trivial)	0.06 (trivial)	0.003 (trivial)	0.22 (small)
<i>P</i>	.160	.496	.095	.188	.435	.312	.458	.188
S1								
Placebo	12.0 (13.5)	28.6 (60.4)	36.9 (14.3)	150.5 (13.5)	21.6 (13.3)	33.8 (58.6)	49.4 (16.2)	151.6 (23.6)
Menthol	9.4 (10.4)	19.6 (57.4)	33.4 (13.5)	142.3 (18.3)	20.8 (8.6)	36.5 (56.6)	49.4 (12.0)	157.2 (23.1)
Standardized difference	-0.21 (small)	-0.15 (trivial)	-0.25 (small)	-0.51 (small)	-0.08 (trivial)	0.05 (trivial)	0.00 (trivial)	0.24 (small)
<i>P</i>	.246	.279	.223	.150	.441	.428	.498	.241
S2								
Placebo	10.2 (10.0)	22.5 (39.1)	29.3 (10.8)	131.1 (13.8)	14.3 (7.5)	23.8 (39.7)	38.5 (10.3)	139.5 (16.7)
Menthol	10.3 (9.2)	18.7 (36.9)	27.1 (13.6)	128.6 (15.0)	20.2 (11.3)	31.7 (45.4)	42.0 (11.2)	141.8 (18.8)
Standardized difference	0.01 (trivial)	-0.10 (trivial)	-0.18 (trivial)	-0.18 (trivial)	0.61 (large)	0.18 (trivial)	0.32 (small)	0.130 (trivial)
Adjusted <i>P</i>	.493	.335	.337	.336	.093	.212	.203	.302

Abbreviations: HiMets, high metabolic load; HSR, high-speed running (>7.5 m·s⁻¹); S1, swill 1; S2, swill 2; Sprint, sprint distance (>5.5 m·s⁻¹). Values are represented as mean (SD).

Bold values are statistically significant.

Hydration Data

No significant differences ($P > .05$) were observed in fluid intake, sweat loss, and sweat rate between the trials for the Forwards or the Backs ($P > .05$). There was a significantly greater percentage dehydration ($P = .04$) among the Backs on the MR trial (Table 4).

Discussion

The primary observation was that a 0.1% L-menthol MR had no effect on selected performance metrics in Super Rugby players (both Forwards and Backs), which rejects our hypothesis that it would benefit rugby-specific conditioning. This contrasts previous research, where MR has shown beneficial performance improvements and attenuations in TS in active participants during endurance type exercise: such as cycling time to exhaustion,¹⁹ 5-km running time trials,²⁶ and both running and cycling time trial outdoors.³¹ In line with our second hypothesis MR did, significantly attenuate the rise in TS among Forwards but not Backs, despite no performance improvements.

Within the population and exercise task of the present study, attenuation of TS alone may not be enough to favorably alter performance. If longer conditioning blocks were utilized and/or match play, then a performance improvement may have been seen; however, this requires further investigation. Other research has found similar relationships during a 40-minute cycling intermittent sprint protocol, with no differences in peak power and total work done despite improved thermal comfort scores with the MR compared with placebo.²⁵ The difference in findings between high-intensity exercise and endurance protocols may be due to differences in the etiology of fatigue between these activities, meaning a perceptual intervention is less effective.²⁵ A physiological response such as a decrease in core and/or skin temperature may be required for a positive performance benefit in this type of exercise and environmental conditions,³² rather than just a change in the perceptual response. This could be a contributing factor to why the benefits in endurance-based studies look different to intermittent-type activities, like team sports. Additionally, environmental conditions could have played a role in the outcomes in this present study, as the ambient conditions reported may not have been sufficient to elicit a strong effect with a MR. Previous research in laboratory conditions had participants exercising in temperatures ≥ 30 °C^{19,25,26} compared with 23 °C in the present study. However, participants did confirm that they “were very hot” during the team training and at the very least they were experiencing perceptual heat strain (which MR has efficacy to favorably alter) during exercise. In accordance, the attenuation of TS through the mechanisms of a MR may need to be greater than observed in the present study to elicit behavioral changes.²⁸

In contrast to the significant differences in TS seen in Forwards, none were seen Backs. Differences in workload, positional patterns, and body composition between Forwards and Backs^{1,2,33} could contribute to the differential responses seen in TS. The average body weight and %body fat was higher in the Forwards (9.7% [1.6%]) compared with Backs (7.7% [0.9%]). Those with a higher body fat percentage (and weight) such as the Forwards, tend to be less effective at dissipating heat and are more susceptible to heat stress than those who have greater lean mass (and are lighter) such as the Backs.¹⁴ This theory to a certain degree is supported by the association between %body fat and change in TS in the present study. Given, the Forwards had a greater average TS score in the PLA trial for both the baseline to S1 (Forwards 1.7–2.9, Backs 2.6–2.75) and baseline to S2 (Forwards 1.7–3.7, Backs 2.6–2.9) compared with the Backs (Figure 2A and 2B), this could indicate a higher thermal strain. However, T_{re} was not investigated in this present study to examine the thermoregulatory differences between positions.

Forwards and Backs “liked” (significantly) the sensation of the MR over PLA at baseline. Although, this preference remained at S3 in the forwards only. Interestingly, although we did not measure acceptability after each swill due to time constraints of the training, some players stated that “it gets worse the more I have.” Therefore, the repeated dosing strategy of a MR in elite rugby players requires further investigation to determine optimal timing of MR. The MR concentration of 0.1% was developed based on preliminary acceptability testing from direct player feedback. Other studies have demonstrated the use of lower concentration L-menthol 0.01% solutions.^{19,25,26} However, research has shown higher menthol concentrations such as 0.1% (as used in this study) could be an “optimal” concentration.³⁴ Given the differences in the “liking” of the MR in the present study, as well as previous research noting the need for liking of an intervention for it to be effective,²⁴ it would seem prudent for the “optimal” concentration of menthol to be found within the specific population it is intended to be used within. It may also be worth noting the use of a light blue or green solution may be prudent in future research in order to potentially further enhance perceptual cooling benefits, as suggested in Best et al.²²

Menthol use has been hypothesized to alleviate the drive to drink due to its interaction with sensory cold receptors in the oropharynx.³⁵ However, there were no differences in fluid intake for MR compared with the PLA group among the backs or the forwards. This result comes despite menthol potentially conferring hedonic and thirst attenuating responses.³⁶ The lack of a significant difference in fluid intakes could be attributed to the cold water PLA swill, which may have elicited similar effects to that of the MR.³⁵ Sweating (sweat loss and sweat rate) was not affected, which is similar to the findings of Gibson et al²⁵ who found no significant differences in whole body sweat rate between placebo and menthol groups when a MR was used. However, the nonstatistically significant differences in sweat loss and fluid intake between MR and PLA were accompanied by

Table 4 Hydration Data, Mean (SD)

	Forwards (n = 15)				Backs (n = 12)			
	Placebo	Menthol	Standardized difference	P	Placebo	Menthol	Standardized difference	P
% dehydration	1.18 (0.61)	1.14 (0.56)	−0.06 trivial	.40	1.13 (0.67)	1.36 (0.54)	0.37 small	.04
Fluid intake, mL	1442 (518)	1614 (584)	0.31 small	.13	1116 (452)	1068 (428)	−0.11 trivial	.37
Sweat loss, L	2.85 (0.72)	2.92 (0.57)	0.11 trivial	.33	2.22 (0.39)	2.28 (0.52)	0.12 trivial	.37
Sweat rate, L·h ^{−1}	1.32 (0.34)	1.36 (0.26)	0.11 trivial	.34	1.03 (0.18)	1.07 (0.23)	0.17 trivial	.32

Note. Significant difference between menthol and placebo represented as $P < .05$.

significantly different % dehydration during the sessions, although the difference is small and unlikely to be clinically meaningful (0.23%). Although speculative, the increased fluid intake may have been an attempt by players to remove the taste of the MR as the session progressed, due to the waning “liking.” Given the constraints of the current study, for example, duration and ambient temperature this finding should be further investigated in future studies where sweat losses have the potential to be greater, such as in hotter more humid conditions with greater exercise durations.

Limitations

The uncontrolled environment and conducting research during an elite, real-world team training session could be considered a limitation. However, it is also strength as it provides ecological validity in a real-world elite team sport setting, an experimental setting with a paucity of data. Given the waning “liking” of MR from baseline to posttraining, future work should investigate the optimal timing of MR before and during exercise.

Practical Applications

1. The findings provide a key starting point for the possible future use of menthol in team sports.
2. This study provides evidence that MR at 3-minute intervals is too short in elite rugby players and produces waning “liking,” with some players stating “it got worse the more I had.”
3. The use of a MR may be more relevant for Forwards than Backs by attenuating the rise in TS. Although individual variation will occur, and application should be individualized.

Conclusion

The data show that MR had no significant effect on any selected performance metrics but did significantly attenuate the rise in TS among Forwards. Additionally, it was significantly more liked at baseline than a water placebo, but liking waned over time.

Acknowledgments

This study was funded by the Department of Human Nutrition, University of Otago. The study concept and designed involved Black, Baker, Taylor, Smith, and Healey. Material preparation, data collection, and analysis were performed by Jerram, Black, Baker, Smith, and Healey. The first draft of the manuscript was written by Jerram, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

References

1. Duthie G, Pyne D, Hooper S. Applied physiology and game analysis of rugby union. *Sports Med.* 2003;33(13):973–991. doi:10.2165/00007256-200333130-00003
2. Deutsch MU, Kearney GA, Rehrer NJ. Time - motion analysis of professional rugby union players during match-play. *J Sports Sci.* 2007;25(4):461–472. doi:10.1080/02640410600631298
3. Cunniffe B, Proctor W, Baker JS, Davies B. An evaluation of the physiological demands of elite rugby union using global positioning system tracking software. *J Strength Cond Res.* 2009;23(4):1195–1203. doi:10.1519/JSC.0b013e3181a3928b
4. Girard O, Brocherie F, Bishop DJ. Sprint performance under heat stress: a review. *Scand J Med Sci Sports.* 2015;25:79–89. doi:10.1111/sms.12437
5. Racinais S, Alonso JM, Coutts AJ, et al. Consensus recommendations on training and competing in the heat. *Br J Sports Med.* 2015;49(18):1164–1173. doi:10.1136/bjsports-2015-094915
6. Wendt D, van Loon LJ, Lichtenbelt WD. Thermoregulation during exercise in the heat: strategies for maintaining health and performance. *Sports Med.* 2007;37(8):669–682. doi:10.2165/00007256-200737080-00002
7. Bongers CC, Hopman MT, Eijssvogels TM. Cooling interventions for athletes: an overview of effectiveness, physiological mechanisms, and practical considerations. *Temperature.* 2017;4(1):60–78. doi:10.1080/23328940.2016.1277003
8. Taylor L, Thornton HR, Lumley N, Stevens CJ. Alterations in core temperature during world rugby sevens series tournaments in temperate and warm environments. *Eur J Sport Sci.* 2019;19(4):432–441. doi:10.1080/17461391.2018.1527949
9. Henderson MJ, Christmas BCR, Stevens CJ, Coutts AJ, Taylor L. Changes in core temperature during an elite female rugby sevens tournament. *Int J Sports Physiol Perform.* 2020;15(4):571–580. doi:10.1123/ijssp.2019-0375
10. Black KE, Black AD, Baker D, Fairbairn K. Body mass changes during training in elite rugby union: is a single test of hydration indices reliable? *Eur J Sport Sci.* 2018;18(8):1049–1057. doi:10.1080/17461391.2018.1470677
11. Meir RA, Brooks LO, Rogerson S. What do changes in prematch vs. postmatch, 1, 2, and 3 days postmatch body weight tell us about fluid status in English premiership rugby union players? *J Strength Cond Res.* 2011;25(8):2337–2343. doi:10.1519/JSC.0b013e3181f15945
12. Morrissey MC, Wu Y, Zuk EF, Livingston J, Casa DJ, Pescatello LS. The impact of body fat on thermoregulation during exercise in the heat: a systematic review and meta-analysis. *J Sci Med Sport.* 2021;24(8):843–850. doi:10.1016/j.jsams.2021.06.004
13. Smart DJ, Hopkins WG, Gill ND. Differences and changes in the physical characteristics of professional and amateur rugby union players. *J Strength Cond Res.* 2013;27(11):3033–3044. doi:10.1519/JSC.0b013e31828c26d3
14. Davis JK, Baker LB, Barnes K, Ungaro C, Stofan J. Thermoregulation, fluid balance, and sweat losses in American football players. *Sports Med.* 2016;46(10):1391–1405. doi:10.1007/s40279-016-0527-8
15. Dervis SCG, Chaseling GK, Filingeri D, Smoljanic J, Jay O. A comparison of thermoregulatory responses to exercise between mass-matched groups with large differences in body fat. *J Appl Physiol.* 2016;120(6):615–623. doi:10.1152/jappphysiol.00906.2015
16. Taylor L, Carter S, Stellingwerff T. Cooling at Tokyo 2020: the why and how for endurance and team sport athletes. *Br J Sports Med.* 2020;54(21):1243–1245. doi:10.1136/bjsports-2020-102638
17. Burke LM, Maughan RJ. The Governor has a sweet tooth - mouth sensing of nutrients to enhance sports performance. *Eur J Sport Sci.* 2015;15(1):29–40. doi:10.1080/17461391.2014.971880
18. Anderson EM, Jenkins AC, Caudle RM, Neubert JK. The effects of a co-application of menthol and capsaicin on nociceptive behaviors of the rat on the operant orofacial pain assessment device. *PLoS One.* 2014;9(2):e89137. doi:10.1371/journal.pone.0089137
19. Flood TR, Waldron M, Jeffries O. Oral L-menthol reduces thermal sensation, increases work-rate and extends time to exhaustion, in the heat at a fixed rating of perceived exertion. *Eur J Appl Physiol.* 2017;117(7):1501–1512. doi:10.1007/s00421-017-3645-6
20. Hopkins WG, Marshall SW, Batterham AM, Hanin J. Progressive statistics for studies in sports medicine and exercise science. *Med Sci*

- Sports Exerc.* 2009;41(1):3–12. doi:[10.1249/MSS.0b013e31818cb278](https://doi.org/10.1249/MSS.0b013e31818cb278)
21. Barwood MJ, Gibson OR, Gillis DJ, et al. Menthol as an Ergogenic aid for the Tokyo 2021 Olympic games: an expert-led consensus statement using the modified Delphi method. *Sports Med.* 2020; 50(10):1709–1727. doi:[10.1007/s40279-020-01313-9](https://doi.org/10.1007/s40279-020-01313-9)
 22. Best R, Spears LR, Hurst P, Berger NJA. The development of a menthol solution for use during sport and exercise. *Beverages.* 2018;4(2):44. doi:[10.3390/beverages4020044](https://doi.org/10.3390/beverages4020044)
 23. Eccles R. Menthol and related cooling compounds. *J Pharm Pharmacol.* 1994;46(8):618–630. doi:[10.1111/j.2042-7158.1994.tb03871.x](https://doi.org/10.1111/j.2042-7158.1994.tb03871.x)
 24. Best R, McDonald K, Hurst P, Pickering C. Can taste be ergogenic? *Eur J Nutr.* 2020;60(1):45–54. doi:[10.1007/s00394-020-02274-5](https://doi.org/10.1007/s00394-020-02274-5)
 25. Gibson OR, Wrightson JG, Hayes M. Intermittent sprint performance in the heat is not altered by augmenting thermal perception via L-menthol or capsaicin mouth rinses. *Eur J Appl Physiol.* 2019;119(3): 653–664. doi:[10.1007/s00421-018-4055-0](https://doi.org/10.1007/s00421-018-4055-0)
 26. Stevens CJ, Thoseby B, Sculley DV, Callister R, Taylor L, Dascombe BJ. Running performance and thermal sensation in the heat are improved with menthol mouth rinse but not ice slurry ingestion. *Scand J Med Sci Sports.* 2016;26(10):1209–1216. doi:[10.1111/sms.12555](https://doi.org/10.1111/sms.12555)
 27. Yuhasz MS. *The Effects of Sports Training on Body Fat in Man With Predictions for Optimal Body Weight.* University of Illinois; 1962.
 28. Jeffries O, Goldsmith M, Waldron M. L-Menthol mouth rinse or ice slurry ingestion during the latter stages of exercise in the heat provide a novel stimulus to enhance performance despite elevation in mean body temperature. *Eur J Appl Physiol.* 2018;118(11):2435–2442. doi:[10.1007/s00421-018-3970-4](https://doi.org/10.1007/s00421-018-3970-4)
 29. Mundel T, Jones DA. The effects of swilling an L(-)-menthol solution during exercise in the heat. *Eur J Appl Physiol.* 2010;109(1):59–65. doi:[10.1007/s00421-009-1180-9](https://doi.org/10.1007/s00421-009-1180-9)
 30. Zhang H, Huizenga C, Arens E, et al. Thermal sensation and comfort in transient non-uniform thermal environments. *Eur J Appl Physiol.* 2004;92:728–733. doi:[10.1007/s00421-004-1137-y](https://doi.org/10.1007/s00421-004-1137-y)
 31. Tran Trong T, Riera F, Rinaldi K, Briki W, Hue O. Ingestion of a cold temperature/menthol beverage increases outdoor exercise performance in a hot, humid environment. *PLoS One.* 2015;10(4): e0123815. doi:[10.1371/journal.pone.0123815](https://doi.org/10.1371/journal.pone.0123815)
 32. Gerrett N, Jackson S, Yates J, Thomas G. Ice slurry ingestion does not enhance self-paced intermittent exercise in the heat. *Scand J Med Sci Sports.* 2017;27(11):1202–1212. doi:[10.1111/sms.12744](https://doi.org/10.1111/sms.12744)
 33. Nicholas CW. Anthropometric and physiological characteristics of rugby union football players. *Sports Med.* 1997;23(6):375–396. doi:[10.2165/00007256-199723060-00004](https://doi.org/10.2165/00007256-199723060-00004)
 34. Stevens CJ, Ross MLR, Vogel RM. Development of a “cooling” menthol energy gel for endurance athletes: effect of menthol concentration on acceptability and preferences. *Int J Sport Nutr Exerc Metab.* 2020;31(1):40–45. doi:[10.1123/ijnsnem.2020-0190](https://doi.org/10.1123/ijnsnem.2020-0190)
 35. Eccles R. Role of cold receptors and menthol in thirst, the drive to breathe and arousal. *Appetite.* 2000;34(1):29–35. doi:[10.1006/appe.1999.0291](https://doi.org/10.1006/appe.1999.0291)
 36. Best R, Maulder PS, Berger N. Perceptual and physiological responses to carbohydrate and menthol mouth-swilling solutions: a repeated measures cross-over preliminary trial. *Beverages.* 2021;7(1): Article 9. doi:[10.3390/beverages7010009](https://doi.org/10.3390/beverages7010009)